Test

A
X
Y
a@b.c

1 Method

1.1 Setup: Symmetric Estimation

Let the set of investor types be $\mathcal{I} = \{\text{foreign}, \text{institutional}, \text{retail}\}$. Each type is treated as an agent whose observed daily order flow reveals a latent reward defined over interpretable market state features.

Continuous action. Writing $V_{i,t}^{buy}$ and $V_{i,t}^{sell}$ for the daily gross buy and gross sell value of type i, we define the action as

$$u_{i,t} = \frac{V_{i,t}^{buy} - V_{i,t}^{sell}}{V_{i,t}^{buy} + V_{i,t}^{sell}} \in [-1, 1]. \tag{1}$$

Because the denominator is the type's own gross turnover rather than market-wide turnover, $u_{i,t}$ measures whether buying or selling dominated *within* that type, giving a common scale that is unaffected by differences in trading size across types. Using a continuous intensity rather than a binary buy/sell label preserves the magnitude information that dichotomization discards.

Symmetry. All three types receive the same state features $x_{i,t} \in \mathbb{R}^3$, the same market context $C_t \in \mathbb{R}^2$, and the same model form. No type is assigned a different feature a priori; only the parameters $\theta_i = (\beta_i, B_i, \alpha_i)$ are estimated separately on each type's own data. Any difference that appears in the results is therefore learned from the data rather than imposed by design, and this symmetry is what licenses direct comparison of coefficients across types.

Timing. The training sample is $(x_{i,t}, C_t, a_{i,t+1})$ with $a_{i,t+1} = u_{i,t+1}$. A state built from day-t closing prices and order flow explains day-$t+1$ behaviour, so no future trade enters the features. Momentum and underwater use day-t information while the herding feature uses day $t-1$. Since the target is day-$t+1$ behaviour, day-t information is also available; the extra lag on herding is therefore not an information constraint but a design choice that keeps the same-day accounting offset between types out of the predictor set (Figure 10).

1.2 Reward Features

All three features are standardized using statistics fitted on the training indices of each validation split only. Every coefficient reported below is therefore read as the effect of a one-standard-deviation move of that feature within the training window.

(F1) Medium-term momentum. With adjusted (or fallback) closing price P_t,

$$x_{i,t}^{mom} = \log\left(\frac{P_t}{P_{t-20}}\right). \tag{2}$$

Twenty trading days is a pre-specified window of roughly one month, chosen to damp daily price noise while still expressing a medium-term trend linked to next-day behaviour. It is a design value for interpretability, not a value tuned on the data. A positive coefficient indicates trend following and a negative one contrarian trading; the sign expectation follows prior work on type-specific momentum trading [1, 3].

(F2) Herding. The cross-sectional average of the previous day's actions of the two types other than i:

$$x_{i,t}^{herd} = \frac{u_{j,t-1} + u_{k,t-1}}{2}, \qquad j, k \neq i. \tag{3}$$

Note that this is an average across the two other types at one point in time, not a time average. A 20-day moving average would measure a month of cumulative flow pressure rather than an immediate response, so we focus on the previous day. A positive coefficient means moving with the previous day's direction of the other types, a negative one means moving against it. The sign expectation follows the institutional herding literature [6]; note however that this literature measures herding *within* the institutional group, whereas Equation (3) measures a relation *between* types (§2.7).

(F3) Underwater gap relative to average cost. With a per-type average-cost proxy $\bar{P}_{i,t}$ and the day's gross-trade VWAP $P_{i,t}^{vwap}$,

$$x_{i,t}^{uw} = \max\left(\frac{\bar{P}_{i,t} - P_{i,t}^{vwap}}{\bar{P}_{i,t}}, 0\right). \tag{4}$$

Truncation at zero makes the feature active only when the traded price sits below the average-cost proxy, so the variable is reference-point dependent by construction. The average cost is obtained by recursively updating an effective inventory from gross buy and sell quantities and values: the previous day's inventory is multiplied by a forgetting factor $\rho = 0.98$ to damp the influence of old trades, and sales do not change the average cost of the remaining position. With $\rho = 0.98$ the half-life is about 34 trading days and roughly 66.8% remains after 20, so the feature emphasises the trading memory of the past one to two months; this value was also not tuned on the data. A positive coefficient means stronger next-day buying below the average cost, and the sign expectation follows the disposition-effect literature [7, 8].

The proxy is constructed from aggregate flow rather than actual account holdings. Buys and sells executed by different accounts within the same type net out, so the notion of a "representative investor's average cost" is itself an approximation. The coefficient on x^{uw} therefore cannot be read as a direct test of the disposition effect.

1.3 Market Context

The context consists of two variables: (C1) the one-day KOSPI200 return r_t, and (C2) the KRW/USD exchange rate level

$$z_t = \frac{\log e_t - \mu_{t-1,252}}{\sigma_{t-1,252}}, \tag{5}$$

where the current rate is used but the mean and standard deviation are computed only through $t-1$ over 252 trading days. Both contexts are likewise re-standardized on the training indices of each split, so their coefficients read as the effect of a one-standard-deviation move within the training window.

Context and reward features are not independent. Because Samsung Electronics carries a large weight in the KOSPI200, stock momentum overlaps with the market return, and feature–context correlations range from -0.29 to 0.22 (Figure 9). Individual coefficients are therefore conditional associations given the other terms and should not be read as causal magnitudes in isolation.

1.4 Myopic Reward, Policy, and the Reduction to Lasso

Latent action score. Define the action score of type i at time t as

$$q_{i,t} = (\beta_i + B_i C_t)^\top x_{i,t} + \alpha_i^\top C_t, \tag{6}$$

where $\beta_i \in \mathbb{R}^3$ is the baseline reward weight at average context, $B_i \in \mathbb{R}^{3\times 2}$ the feature–context interaction, and $\alpha_i \in \mathbb{R}^2$ the direct context main effect. The α term is a level effect that shifts behaviour even when the stock features are at their means; the B term is a slope effect that changes sensitivity to each feature across market conditions. Including interaction terms while omitting the lower-order term C_t is a standard specification error, so the inclusion of α is required by the specification rather than something to be justified empirically.

Myopic concave reward. The one-step reward for a continuous action a is

$$R_i(x, C, a) = q_{i,t}\, a - \frac{\kappa}{2} a^2. \tag{7}$$

The quadratic term imposes a symmetric cost on large net buying and net selling and makes the reward concave in the action, which permits an interior solution. Without it, maximizing over $[-1, 1]$ would return ± 1 almost always depending only on the sign of q, and the *intensity* of behaviour could not be reproduced. Maximizing Equation (7) over a yields the closed-form policy

$$a^*_{i,t+1} = \operatorname{clip}\left(\frac{q_{i,t}}{\kappa}, -1, 1\right). \tag{8}$$

The curvature κ is fixed at one for identification. With $\kappa = 2$ the same q would halve the action size, but re-estimation would double the coefficients of q and reproduce the same behaviour; $\kappa = 1$ therefore fixes the units of the reward and the coefficients rather than asserting that the empirical curvature equals one.

Estimation. The training objective for each type on its training set $\mathcal{D}_i$ is

$$\min_{\theta_i} \frac{1}{|\mathcal{D}_i|} \sum_{t \in \mathcal{D}_i} \left(a^*_{i,t+1} - a_{i,t+1}\right)^2 + \lambda_i \|\theta_i\|_1, \tag{9}$$

optimized independently per type with Adam [5].

Proposition 1.1 (Reduction to Lasso). *Let* $z_{i,t} = [\,x_{i,t};\ \operatorname{vec}(x_{i,t}C_t^\top);\ C_t\,] \in \mathbb{R}^{11}$ *and* $\theta_i = [\,\beta_i;\ \operatorname{vec}(B_i);\ \alpha_i\,]$, *so that Equation* (6) *reads* $q_{i,t} = \theta_i^\top z_{i,t}$. *If* $|q_{i,t}| \le 1$ *throughout the training sample, the clip in Equation* (8) *never binds,* $a^* = q$, *and Equation* (9) *becomes*

$$\min_{\theta_i} \frac{1}{|\mathcal{D}_i|} \sum_t \left(\theta_i^\top z_{i,t} - a_{i,t+1}\right)^2 + \lambda_i \|\theta_i\|_1, \tag{10}$$

which is exactly an L_1*-penalized (Lasso) regression of next-day net-buying intensity on the expanded state feature* z.

Under a myopic concave reward with $\kappa = 1$, "recovering a preference" and "regressing order flow" are therefore numerically indistinguishable. Figure 8 shows that the premise of Proposition 1.1 in fact holds in our sample: for all three types the distribution of the latent action score sits far inside the action bounds ± 1 (the maximum $|q|$ is 0.842 for foreign, 0.587 for retail and 0.135 for institutional investors), and the saturation rate is exactly zero in all 45 splits.

Why we do not hide the reduction. This coincidence is not a weakness of the method but a logical consequence of the myopia assumption. If the agent looks only one step ahead, today's action does not affect anything beyond tomorrow, so there is no future to look toward and *what the agent wants* (the reward) collapses onto *what the agent does* (the policy). Applying the stochastic policy of maximum-entropy IRL [9], $\pi(a \mid x, C) \propto \exp(R/\tau)$, to Equation (7) makes π a truncated normal with mean q and variance τ, whose maximum-likelihood point estimate coincides with Equation (10). The reduction is thus a property of the myopic, quadratic-reward setting in general, not an artefact of one implementation.

The distinctive value of the inverse reinforcement learning formulation lies in the extension path. Once state dynamics and multi-period value are introduced, the reward separates from the policy and leads to a structural recovery that regression cannot reach. This paper is the first step on that path, and it adds interpretation and validation on top of an explicitly stated set of conditions.

1.5 Validation Protocol

Rather than stopping at the sign and magnitude of the recovered coefficients, we apply four stages.

(V1) Out-of-sample stability under CPCV We use combinatorial purged cross-validation [2] with ten temporal folds of which two serve as test folds, giving $\binom{10}{2} = 45$ splits, a one-day purge and a five-day embargo between training and test, and all standardization fitted on training indices only. We report the fraction of splits in which the dominant sign appears (sign consistency), but because the splits share observations we interpret sign consistency only as a stability indicator, never as an independent-sample p-value.

(V2) Calendar-month block bootstrap To address the fact that sign consistency is not a test, we partition the sample into calendar-month blocks, resample months with replacement, and re-estimate 1,000 times to obtain 95% confidence intervals. Month blocks preserve the autocorrelation of daily flow and within-month patterns inside each block.

(V3) Expanding walk-forward Because CPCV can place a test fold before a training fold, we add a validation that strictly respects the arrow of time. We train on the first 400 observations, predict the next 60 trading days after a five-day purge, and repeat while expanding the training window by 60 trading days, for ten windows in total.

(V4) Regularization swap To confirm that the results are not an artefact of the L_1 penalty, we re-run the same protocol with an L_2 penalty and with no penalty, and compare coefficient signs.

Metrics. Magnitude error is measured by MAE and RMSE, and directional reproduction by directional accuracy, the fraction of days with $\text{sign}(a^*) = \text{sign}(a)$. We also report the Pearson correlation between actual and predicted actions and the saturation rate, the fraction with $|a^*| = 1$. Because action standard deviations differ across types, absolute MAE and RMSE are not compared across types but interpreted within each type, and we report errors normalized by the action standard deviation alongside them. As a reference we also report a persistence rule that predicts the previous day's own action (§2.8).

Construct validity check. We contrast the recovered signs with expectations derived from independently established behavioural-finance regularities (§2.7). This is a comparison against sign expectations fixed in advance, not a post hoc interpretation.

2 Experiments

2.1 Data and Setup

The raw data are daily records for Samsung Electronics (005930) and include prices, trading value, gross buy and sell values and quantities by investor type, the KRW/USD exchange rate, and KOSPI200 returns. Among the observations for which the 252-day exchange-rate history, the 20-day momentum, and the next day's action can all be computed, we use the most recent 973 rows. State dates run from 6 January 2022 to 29 December 2025, with the last state targeting the action of 30 December 2025.

The single-stock design is not a limitation of the sample but a controlled demonstration setting that isolates preference heterogeneity across types without cross-stock confounding. Because stock fixed effects, liquidity differences, and index-inclusion effects do not intervene, differences in coefficients across types can be attributed to learning rather than design.

Table 1: Data and estimation setup

Item	Setting
Sample	973 trading days, 2022-01-06–2025-12-29
Types	foreign, institutional, retail (symmetric estimation)
Reward features	momentum (20d), herding (1d lag), underwater ($\rho = 0.98$)
Contexts	KOSPI200 1d return, USD/KRW 252d level z
Parameters per type	11 (β 3, B 6, α 2)
Common settings	seed 42, CPU, $\kappa = 1$, train-only scaling
Foreign	75 epochs, batch 256, lr 0.001, $\lambda = 0$
Institutional	10 epochs, batch 2048, lr 0.0005, $\lambda = 0.01$
Retail	20 epochs, batch 512, lr 0.001, $\lambda = 0.0003$
Validation	CPCV 10 folds, 2 test folds, purge 1, embargo 5, 45 splits
Stability	month-block bootstrap 1,000, walk-forward 10 windows, penalty swap

Mean actions are -0.0063 for foreign, -0.0100 for institutional and 0.0035 for retail investors, all close to zero, but standard deviations differ markedly: retail is largest at 0.334 and institutional smallest at 0.125. First-order autocorrelations are 0.401, 0.149 and 0.345 respectively.

Same-day actions across types are strongly negatively related (Figure 10). The foreign–retail correlation is -0.772 and the institutional–retail correlation is -0.499, reflecting the market-clearing constraint that one type's buying must appear as another's selling. This fact is used again to bound the interpretation in §2.3 and §2.4.

2.2 Out-of-Sample Behaviour Reproduction

Figure 1 places actual and predicted net buying side by side for the three types. Three things are immediately visible. First, for foreign and retail investors the predicted series tracks the regime turns of the actual series, with smoothed correlations of 0.69 and 0.76. Second, the predicted amplitude is markedly smaller than the actual one, a consequence of coefficient shrinkage under the L_1 penalty. Third, the institutional prediction is essentially pinned to zero (0.02): the null reported in §2.6 appears first in the prediction series rather than in a coefficient table.

Table 2 and Figure 2 reveal two things that means alone conceal. First, dispersion across splits is large: the foreign correlation ranges from -0.05 to 0.45, so comparing types from a single split is meaningless. Second, the naive persistence baseline outperforms the main model on both directional accuracy and correlation. Since a_t is published after the day-t close, this is an informationally feasible rule. The

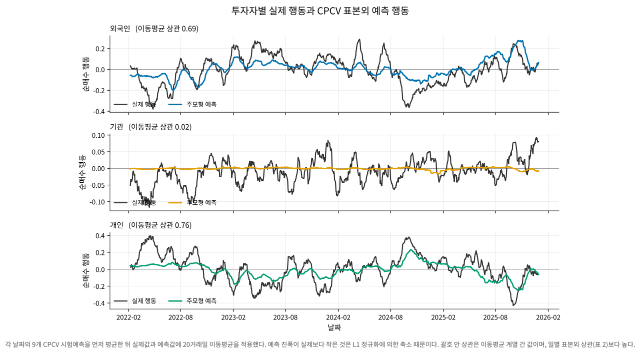

Figure 1: Actual and out-of-sample predicted net buying by investor type. The nine CPCV test predictions available for each date are averaged first, then a 20-day moving average is applied to both the actual and the predicted series. Correlations in the panel titles are between the smoothed series and are not directly comparable to the daily out-of-sample correlations in Table 2.

Table 2: Out-of-sample performance under CPCV (mean ± s.d. over 45 splits) and the persistence baseline

Metric	Foreign	Institutional	Retail
Main model			
Directional accuracy	0.636 ± 0.039	0.542 ± 0.038	0.608 ± 0.044
Correlation	0.278 ± 0.110	0.068 ± 0.064	0.245 ± 0.118
MAE	0.195 ± 0.019	0.094 ± 0.008	0.264 ± 0.029
RMSE	0.246 ± 0.021	0.125 ± 0.012	0.318 ± 0.030
MAE / action s.d.	0.759	0.756	0.791
Saturation rate	0.000	0.000	0.000
Persistence baseline ($a_{t+1} \leftarrow a_t$)			
Directional accuracy	**0.642**	**0.547**	**0.616**
Correlation	**0.359**	**0.136**	**0.311**
MAE	0.223	0.123	0.299
Action AR(1)	0.401	0.149	0.345

main model leads only on MAE, and because that follows from shrinkage of the predicted amplitude it cannot be read as an informational advantage.

The reason for the gap is clear. The action has first-order autocorrelation of 0.401/0.149/0.345, yet our reward feature set does not include the lagged own action. Restricting the features to (F1)–(F3) was a design choice ensuring that all three types share only features corresponding to independently established behavioural regularities; it was not a design aimed at maximizing predictive performance. We report this result openly in Table 2. Table 2 and Figure 2 are used to show that the model has not merely fitted noise, not as evidence of predictive superiority.

Because the saturation rate is exactly zero for every type in every split, the premise $|q| \leq 1$ of Proposition 1.1 holds

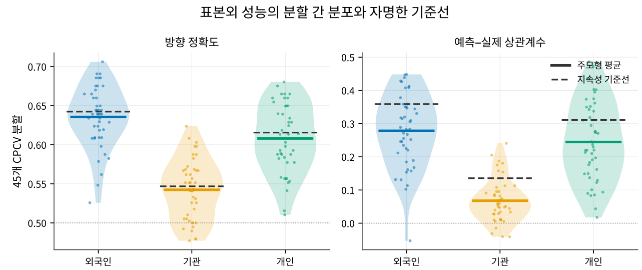

Figure 2: Split-level distribution of out-of-sample performance against a naive baseline. Solid lines are the main-model mean and dashed lines the persistence baseline. The dashed line lies above the solid line for all three types.

throughout the sample. The estimator is therefore literally the Lasso of Equation (10), and performance does not rely on extreme ± 1 predictions.

2.3 Recovered Rewards: A Mirror Image on the Momentum Axis

Figure 3 places the recovered baseline reward weights of the three types side by side. Because all types received the same feature set and the same model form and only their parameters were estimated separately, the differences in bar height are entirely learned from the data. Figure 4 unfolds the same β together with the context main effects α down to the split level.

The mirror image on the momentum axis. The foreign momentum weight is $+0.0496$ and positive in all 45 splits, while the retail weight is -0.0304 and negative in all 45 splits. In Figure 3 the two bars point in opposite directions across the zero line, and in Figure 4 their bootstrap intervals do not overlap and each excludes zero. Although the same feature set was estimated independently for each type, the two types diverge in exactly opposite directions on momentum: foreign investors appear as trend followers and retail investors as contrarians, matching the sign expectations of prior work [1, 3].

Herding. The herding coefficient is negative in all 45 splits for all three types, and the bootstrap intervals of the foreign (-0.0402) and retail (-0.0322) coefficients also exclude zero. This result, however, does not separate a behavioural interpretation from the market-clearing constraint. As Figure 10 shows, the same-day correlation between types reaches -0.77, so a substantial part of the negative herding coefficient may arise mechanically. Because Equation (3) uses a one-day lag the same-day constraint does not apply directly, but it can propagate through the autocorrelation of the action. We therefore restrict this result to the descriptive statement that observed aggregate flow displays a negative cross-type association, and do not extend it to the behavioural claim that each type independently trades against the others.

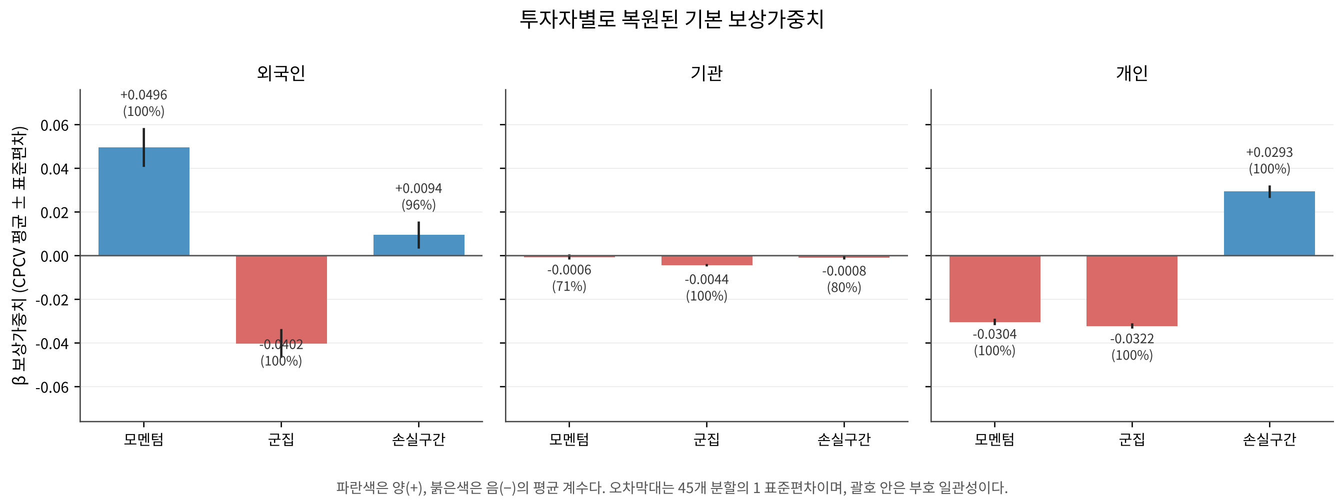

Figure 3: Recovered baseline reward weights β by investor type. Blue bars are positive and red bars negative mean coefficients; error bars are one standard deviation across the 45 splits and the parenthesised figure is sign consistency.

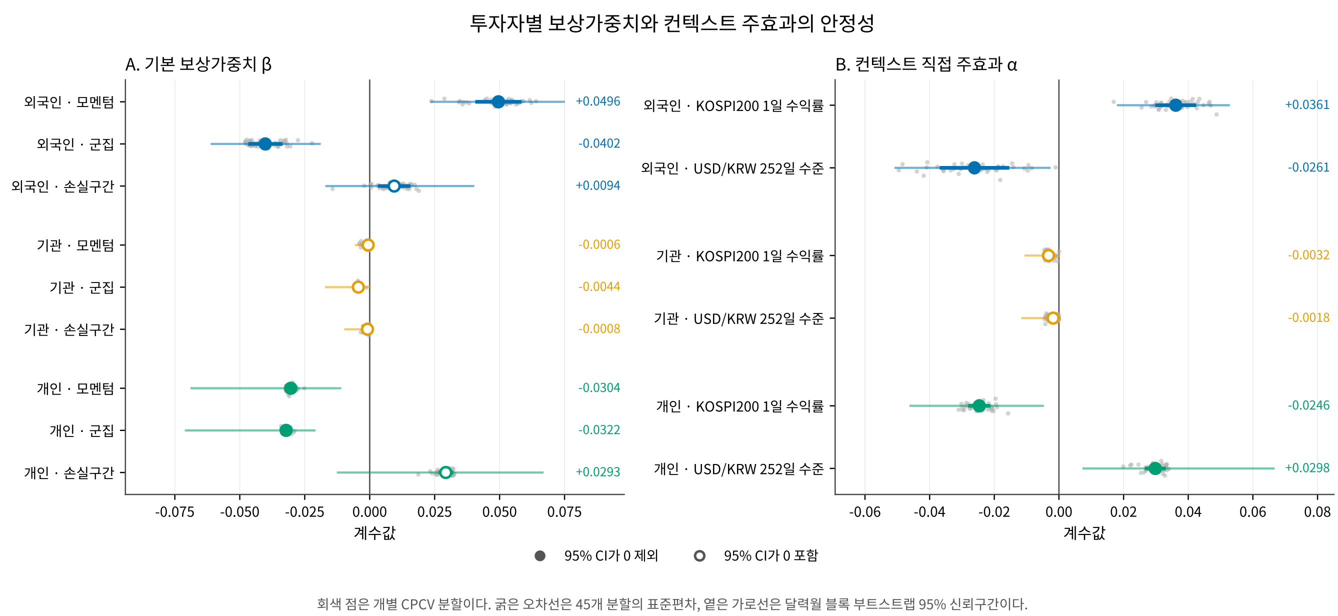

Figure 4: Split-level stability of reward weights and context main effects. Grey dots are individual CPCV splits, thick error bars the standard deviation across the 45 splits, and light horizontal lines the 95% calendar-month block bootstrap interval. Hollow markers denote terms whose interval contains zero.

Underwater. The retail underwater weight is $+0.0293$ and positive in all 45 splits: buying intensity rises on the following day when the traded price is below the average-cost proxy, consistent with the reference-price dependence of the disposition-effect literature [7, 8]. The foreign coefficient shares the sign at $+0.0094$ (95.6%) but is about a third of the retail magnitude.

As Figure 4 shows, however, the underwater bootstrap interval contains zero for all three types. The retail interval is $[-0.0122, +0.0665]$: although the sign is 100% consistent across the 45 splits, it flips in 5.5% of month-block resamples. This is exactly the situation anticipated in §1.5, where sign consistency understates uncertainty because CPCV splits share observations. We therefore report that a sign consistent with the disposition effect was observed, but do not claim it as a statistically established result.

2.4 Context Response: The Mirror Image Reappears

The mirror image observed on the momentum axis reappears in the context main effects (panel B of Figure 4). Foreign investors shift toward net buying on days when the market rises (+0.0361) and toward net selling when the won is weak (−0.0261), while retail investors do exactly the opposite on both contexts (−0.0246 and +0.0298). All four coefficients keep their sign in all 45 splits and all four bootstrap intervals exclude zero. The two institutional coefficients have sign consistency above 95% but are about an eighth of the foreign and retail magnitudes, and their intervals contain zero.

This mirror image requires the same interpretive reservation as the herding coefficient. Given the −0.772 correlation in Figure 10, two types responding to context with opposite signs may in part reflect the market-clearing constraint. We therefore present this as the descriptive fact that the two types form opposite sides of order flow across market and exchange rate conditions.

Interactions. Of the twelve interaction coefficients, four keep their sign in all 45 splits: foreign momentum × KOSPI200 (−0.0139), retail momentum × FX (+0.0144), retail herding × KOSPI200 (−0.0080) and retail underwater × FX (−0.0186). Only the first and the last of these pass the bootstrap. Since twelve coefficients were examined jointly, we present the interactions as exploratory results only.

2.5 Results of the Stability Protocol

(V2) Calendar-month block bootstrap. The light horizontal lines in Figure 4 give this result. The bootstrap is a stricter criterion than CPCV sign consistency, and the outcome splits three ways. First, the momentum mirror and the context mirror survive: eight terms, comprising foreign and retail momentum, herding, and both context main effects, all exclude zero. Second, the underwater weights do not survive. Third, no institutional term excludes zero; all six institutional markers in Figure 4 are hollow.

(V3) Expanding walk-forward. Figure 5 gives this result, and two things read at once. Performance fluctuates sharply across windows: foreign directional accuracy moves between 0.43 and 0.85, averaging 0.610 ± 0.116, and the correlation falls to 0.131 ± 0.162, less than half the CPCV value of 0.278; retail falls from 0.245 to 0.129. Because CPCV can place a test fold ahead of a training fold while walk-forward cannot, we report this degradation as a generalization limit arising from the temporal structure.

The coefficient paths, by contrast, do not fluctuate. The foreign momentum coefficient is positive in all ten windows and the retail one negative in all ten, and the two paths never cross. The same mirror image persists across all windows for both context main effects, while the three institutional paths stay pinned near zero. Performance moves with time; the direction of the recovered preference does not.

(V4) Regularization swap. Figure 6 gives this result. The foreign and retail coefficients are effectively invariant to the choice of penalty, agreeing to four decimal places, so the momentum mirror is not an artefact of the L_1 penalty. Only the institutional coefficients move with the penalty (herding $-0.0076 \to -0.0124$, underwater $-0.0000 \to -0.0018$), on a y-axis range about one twentieth of that of the other two types. That the coefficients depend on the choice of penalty means the data do not determine their value.

2.6 Institutional Investors: A Converged Null

For institutional order flow we find no preference that is stably identified at daily frequency, and we report this as a finding rather than concealing it. The evidence is fivefold and all of it is visible in the figures.

1. **Degenerate prediction.** In the middle panel of Figure 1 the institutional prediction stays on the zero line while the actual series moves between ±0.1.
2. **Magnitude.** In Figure 3 the three institutional bars are barely visible on the shared axis. Excluding herding, momentum is −0.0006 and underwater −0.0008, about one fortieth of the other types.
3. **Uncertainty.** All six institutional markers in Figure 4 are hollow.
4. **Penalty dependence.** In the middle right panel of Figure 6, only the institutional coefficients move with the penalty.
5. **Temporal instability.** In Figure 5 the institutional paths hug the zero line; momentum sign consistency is 40% of ten windows and underwater 0%.

This null is not an optimization failure. Figure 7 shows the training curves across the 45 splits. The institutional training MSE falls by only 1.5% over the whole training run and the curve does not look visually flat. That is not underfitting, however, but the fact that there is almost no error left to remove. By Proposition 1.1, Equation (9) is convex with a unique global optimum, and comparing the solution reached with the closed-form optimum of the same objective leaves an institutional gap of only 0.17% on average and 0.48% at most (0.21% for foreign and 0.92% for retail). All three types reach the attainable optimum; only for institutional investors do the coefficients at that optimum sit near zero. The data do not determine an institutional preference; it is not that the optimizer failed to find one.

Economically the null reads naturally. The “institutional” category aggregates pension funds, investment trusts, insurers, private funds, and securities firms, so the flows of agents with different objective functions offset one another. The one consistently negative herding coefficient is compatible with a shared liquidity-provision role on the opposite side of the other types, but because its bootstrap interval sits on the boundary we do not claim it.

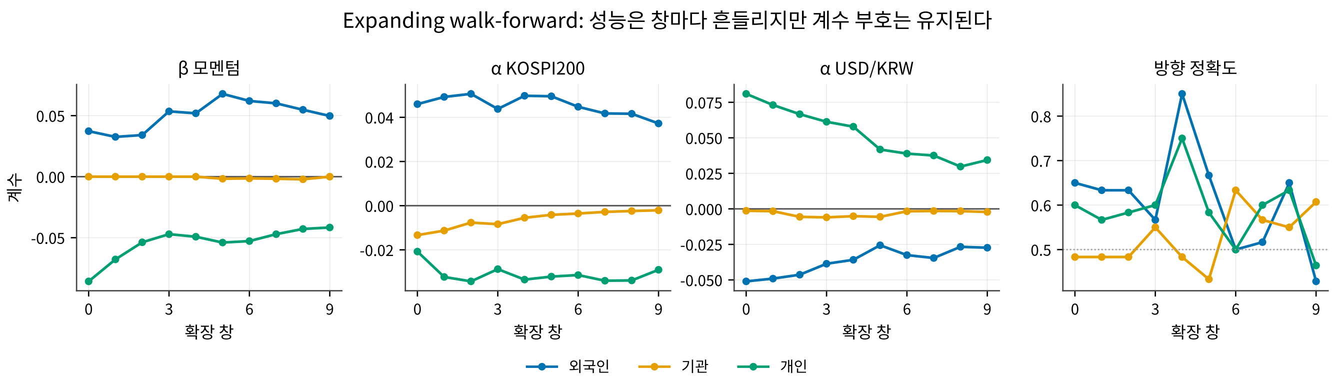

Figure 5: Coefficient paths and directional accuracy over ten expanding walk-forward windows. Performance fluctuates from window to window while the coefficient signs hold.

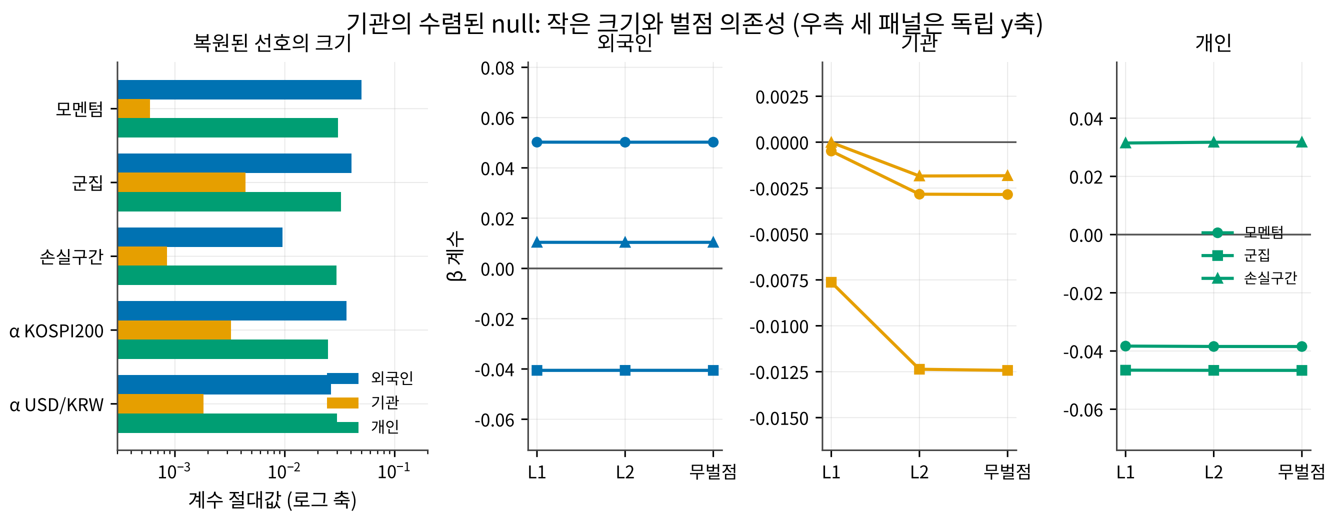

Figure 6: The converged institutional null. Left: magnitude of the recovered preferences on a logarithmic axis. Right: sensitivity to the regularization swap, with an independent y axis in each of the three panels.

Table 3: Prior sign expectations and recovered results

Regularity	Expected	Recovered	Verdict
Foreign trend trading [1, 3]	$\beta^{mom} > 0$	+0.0496 (100%, CI excl. 0)	matches
Retail contrarian trading [3, 4]	$\beta^{mom} < 0$	−0.0304 (100%, CI excl. 0)	matches
Disposition effect [7, 8]	$\beta^{uw} > 0$	+0.0293 (100%, CI incl. 0)	partial
Institutional herding [6]	$\beta^{herd} > 0$	−0.0044 (100%, CI incl. 0)	mismatch

2.7 Construct Validity Check

Table 3 reports the contrast. Although all three types received the same features and the same model form and only their parameters were estimated separately, the recovered signs agree with independently established regularities and the two types diverge in opposite directions on momentum. This is evidence of construct validity: the recovered reward is not an arbitrary fit.

The mismatch arises because the measured object differs. Lakonishok et al. [6] measure the extent to which agents *within* the institutional group move in the same direction, whereas Equation (3) measures whether institutions move with the previous day's flow of the *other two types*. Our result is therefore not a refutation of that literature; testing within-type herding would require data on institutional subcategories.

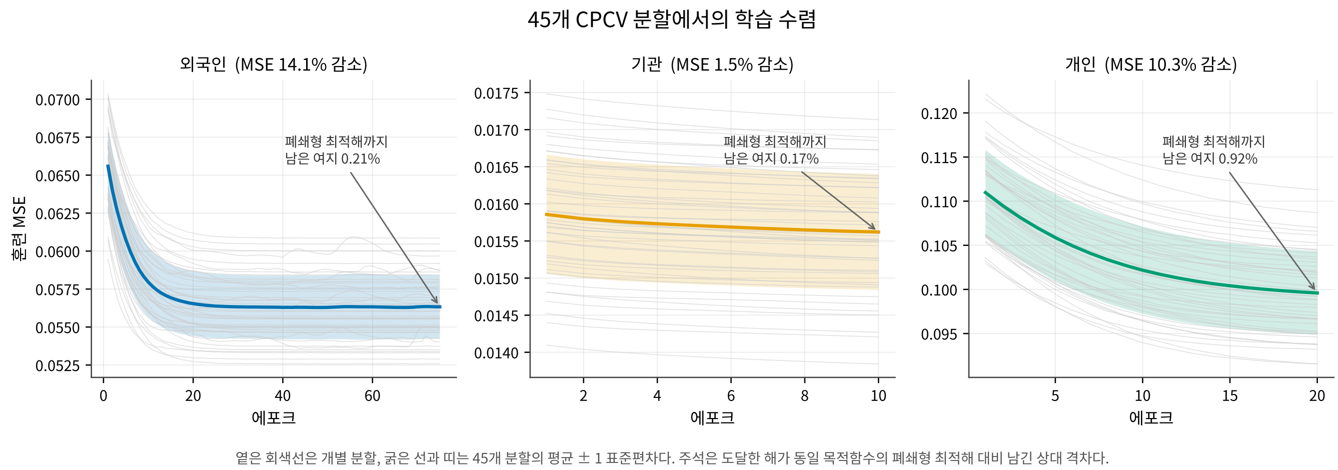

Figure 7: Training convergence across the 45 CPCV splits. Light grey lines are individual splits; the thick line and band are the mean $\pm$ one standard deviation. The annotation gives the relative gap between the solution reached and the closed-form optimum of the same objective.

2.8 What This Study Does Not Claim

As the final stage of the protocol we state explicitly what the results do not support.

(1) **Return predictability.** Our metrics are reproduction errors for the direction and intensity of order flow and include no returns, Sharpe ratios, or transaction costs. No trading rule follows from Table 2.

(2) **Superior flow prediction.** As reported in Table 2 and Figure 2, the persistence baseline beats the main model on directional accuracy and correlation. Our objective is preference recovery on an interpretable common scale rather than predictive performance, and the two objectives call for different feature sets. In a preliminary check, adding the lagged own action as a fourth feature lifts the model above the baseline (directional accuracy 0.646/0.569/0.629) but reverses the sign of the retail α^{KOSPI} and eliminates the underwater coefficient. Some of the mirror images reported here are therefore not separable from the autocorrelation of order flow.

(3) **Causal effects.** The recovered weights are conditional associations among standardized observed features. Because feature–context correlations range from -0.29 to 0.22 (Figure 9), the magnitude of any coefficient depends on the other terms being controlled for.

(4) **A behavioural reading of the negative cross-type association.** The herding coefficient and the mirrored context responses are not separable from the market-clearing constraint (Figure 10).

(5) **A test of the disposition effect.** The retail underwater sign agrees with the literature but does not pass the month-block bootstrap (Figure 4), and the average-cost proxy is built from aggregate trades rather than account-level realized gains and holding periods.

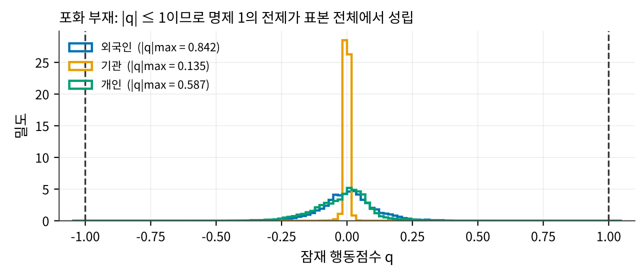

Figure 8: Distribution of the latent action score against the action bounds. Since $|q| \leq 1$, the premise of Proposition 1.1 holds throughout the sample and the saturation rate is exactly zero in all 45 splits.

(6) **Generalization across stocks.** The single-stock design is a control for isolating heterogeneity across types; extension to other stocks and markets is untested.

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Table 4: Interaction coefficients B (45 splits)

Type	Feature × context	Mean	S.d.	Sign cons.
Foreign	momentum × KOSPI200	−0.0139	0.0052	100.0%
Foreign	momentum × FX	−0.0030	0.0057	71.1%
Foreign	herding × KOSPI200	+0.0016	0.0074	66.7%
Foreign	herding × FX	+0.0037	0.0073	68.9%
Foreign	underwater × KOSPI200	−0.0062	0.0040	93.3%
Foreign	underwater × FX	−0.0075	0.0076	86.7%
Inst.	momentum × KOSPI200	−0.0001	0.0005	66.7%
Inst.	momentum × FX	+0.0003	0.0006	80.0%
Inst.	herding × KOSPI200	−0.0004	0.0006	86.7%
Inst.	herding × FX	+0.0012	0.0012	93.3%
Inst.	underwater × KOSPI200	+0.0028	0.0012	97.8%
Inst.	underwater × FX	+0.0007	0.0008	91.1%
Retail	momentum × KOSPI200	+0.0021	0.0042	80.0%
Retail	momentum × FX	+0.0144	0.0045	100.0%
Retail	herding × KOSPI200	−0.0080	0.0046	100.0%
Retail	herding × FX	+0.0092	0.0081	82.2%
Retail	underwater × KOSPI200	−0.0022	0.0046	66.7%
Retail	underwater × FX	−0.0186	0.0046	100.0%

Table 5: Optimality check of the solution reached (45 splits)

Type	MSE drop (training)	Gap (%) mean / max	Cosine mean / min
Foreign	14.1%	0.21/3.10	0.994/0.948
Inst.	1.5%	0.17/0.48	0.935/0.832
Retail	10.3%	0.92/1.40	0.894/0.833

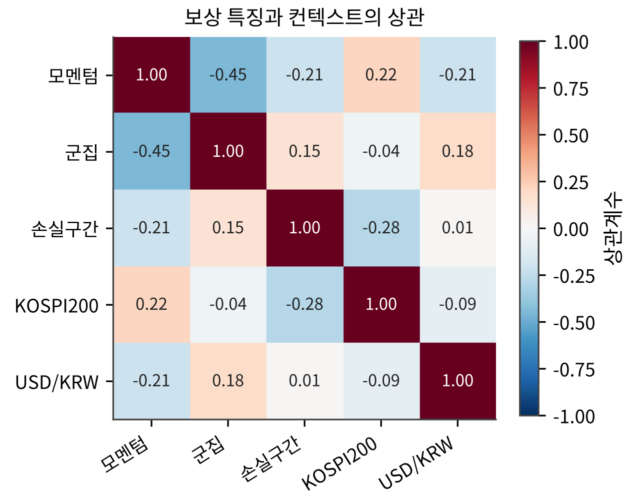

Figure 9: Correlation matrix of reward features and contexts (foreign features, 973 observations).

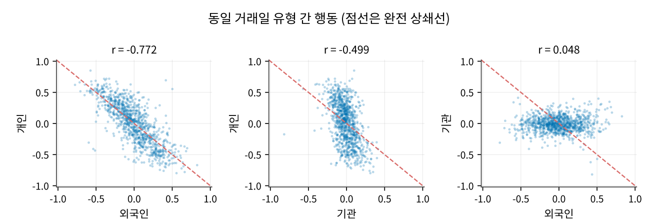

Figure 10: Same-day actions across investor types. The dashed line marks perfect offset; foreign and retail actions cluster along it.

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